Attributions

Team Member Contributions

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Yishuo Zhu

Task: Conceptualization, Background Research, Investigation, Visualisation, Writing

Specific Tasks:

1. Conceptualization: Formulated the primary research objective involving the application of engineered bacterial chassis for the targeted treatment of Small Cell Lung Cancer (SCLC); Proposed and modeled multiple therapeutic signaling pathways and suicide switch to maximize anti-tumor efficacy.
2. Background Research: Conducted extensive systematic reviews of current literature regarding SCLC pathophysiology and the emerging landscape of synthetic biology-based therapeutics.
3. Investigation: Responsible for the targeting and biosafety modules of engineered bacteria, and co-designed the signaling pathways related to quorum sensing (QS) and suicide switches in engineered bacteria.
4. Visualisation: Responsible for schematic illustrations and conceptual visualizations for the Design Part to clearly communicate the synthetic gene circuits and therapies.
5. Writing: Authored the content for the Targeting, Short Peptide Secretion, and Biosafety Modules in the Design part; led the curation of the Registry of Standard Biological Parts and the documentation of the Engineering cycle.

Yuanxin Wang

Task: Conceptualization, Background Research, Investigation, Public Engagement, Safety, Writing, Project Administration

Specific tasks:

1. Conceptualization : participating in brainstorming and developing ideas for the project.
2. Background Research: reading articles on project background and design related topics.
3. Investigation : Searching for evidence to support the choice of chasis organism.

4. Public Engagement: preparing for HP activities (HP, iHP, Education and Cooperation).
5. Safety: Consulting literature to evaluate and confirm the adaptability of EcN in the pulmonary environment.
6. Writing: writing, reviewing and editing content for the Wiki. Writing script for promotion video.
7. Project Administration: managing and recording the project activities.

Duanrui Shen

Task:Conceptualization,Background Research,Writing,Entrepreneurship,Project Administration

Specific tasks:

1.Conceptualization:Actively participated in the initial brainstorming sessions for project topic selection. Analyzed each proposed idea in depth, evaluated feasibility based on literature and synthetic biology principles, and provided constructive feedback.

2.Background Research:Conducted extensive literature review to support project design, particularly in the areas of protein aggregation chemistry, peptide design tools (TANGO, WALZE), nebulization physics, and formulation chemistry.

3.Writing:Undertook the majority of English translation for project documentation, including project descriptions, and technical summaries. Authored and revised the chemistry-related sections (peptide design, formulation, nebulization mechanism) and the entrepreneurship/commercialization analysis.

4.Entrepreneurship:Developed the commercial framework centered on the pulmonary delivery platform and identified competitive advantages of nebulized administration.

5.Project Administration:Leveraging chemistry to advance Project goals.

Sihan Wei

Task: Conceptualization, Writing, Background Research, Public Engagement

Specific tasks:

(1) Conceptualization: Participated in early project brainstorming sessions and contributed to the design of the biosafety module, and provided key input for the foundational ideas of plasmid construction; (2) Writing: Authored and refined the content for the Safety and Contribution pages on the team Wiki, ensuring clarity, accuracy, and compliance with iGEM standards; (3) Background Research: Conducted literature review on small-cell lung cancer (SCLC) and programmed suicide switches, providing theoretical support for the project; (4) Public Engagement: Provided creative inspiration and conceptual support for the Integrated

Human Practices (iHP) and Education outreach activities, helping to shape the team's science communication strategy.

Xiaozheng Han

Tasks: Conceptualization, Background Research, Visualization, Writing.

Specific tasks:

1. Conceptualization: Proposed the aerosolized engineered bacteria delivery for short peptide drugs, integration and construction of a complete genetic circuit including engineered bacteria colonization, quorum sensing and lactate sensing.

2. Background Research: Conducted research on the safety assurance aspects such as pulmonary colonization of engineered bacteria 、QS and lactate-responsive suicide switch.

3. Visualization: Created the schematic diagram of hemolysin secretion system and the first draft of genetic circuit visualization.

4. Writing: Authored, reviewed and edited the project Wiki.

Yingqi Zhou

Tasks: Website Development, Visualization

Specific tasks:

1. **Website Development:** Led the overall construction, design, and maintenance of the team Wiki; developed the site architecture, optimized navigation structure, and ensured technical functionality and responsiveness across different devices and browsers.
2. **Visualization:** Designed the visual identity and layout style of the Wiki, including page templates, color schemes, typography, and interactive elements, to enhance readability and effectively communicate the project's scientific concepts.

Jiacheng Chen

Tasks: Conceptualization, Software, Visualization

Specific tasks:

1. **Conceptualization:** Discuss the possibility and innovativeness of oligopeptide therapy.
2. **Software programming:** Design and program an intergrated and most possibly automated software platform for all the tools used in the prediction of FuncPept, running both on Linux and Windows.

3. **Visualization:** Create some simple animations demonstrating some crucial mechanisms in peptide functioning.

Yuchen Chen

Tasks：Hardware

Specific tasks：

1. **Hardware system development:** Led the overall design of the SCLC hierarchical intelligent atomization terminal, completed SolidWorks 3D modeling, and JLCPCB EDA circuit schematic design
2. **Documentation and Visualization**：Design the structure and presentation format of hardware-related technical documentation, including visual content such as 3D model renderings, exploded views, circuit connection diagrams, and code annotations

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Donglai Liu

Task:

Specific tasks:

Jialin Wu

Task:

Specific tasks:

Project Timeline

1.21 - 1.31 [brainstorming and topic selection]

2.1 - 2.4 [brainstorming and developing the prototype for the topic]

2.9 [safety form, village & age confirmation submission]

2.13 [completing homepage]

2.5 - 2.22 [developing design and engineering]

2.22 [wiki and promotion video submission]